

Photoinduced Enhanced Raman Spectroscopy with Hybrid Au@WS₂ Nanosheets

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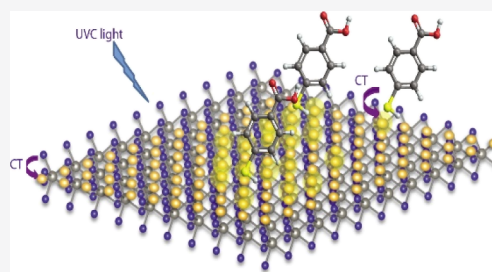


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ABSTRACT: Two-dimensional (2D) layered transition-metal dichalcogenides (2DMX₂) are materials with unique optoelectronic properties, high surface-to-volume ratio, and high carrier mobility. The combination of noble metal nanoparticles (MNPs) with 2DMX₂ opens new avenues in conceiving more efficient plasmonic sensors, allowing one to optimize both electromagnetic and chemical signal enhancement. Photoinduced enhanced Raman spectroscopy (PIERS) exploits the electron migration from semiconductors to MNPs, upon UV light irradiation, to further boost the chemical enhancement in the surface-enhanced Raman scattering (SERS) of molecules deposited on hybrid 2DMX₂–MNP nanostructures. Here, we propose a new PIERS sensor architecture based on tungsten disulfide (WS₂) nanosheets produced by liquid-phase exfoliation (LPE) and functionalized with citrate-stabilized Au MNPs. Electron injection from WS₂ to AuNPs is observed when the Au@WS₂ is exposed to ultraviolet light, yielding an increase of the charge carriers' density $\approx 1.8\%$. The PIERS sensor performances are tested by detecting 4-mercaptobenzoic acid at a concentration of 10 μM . The overall PIERS signal enhancement is $\sim 10^6$, whereas the photoactivation of WS₂ yields a signal improvement of factor 4 with respect to SERS from Au@WS₂ before UV irradiation. Our sensor is of low cost, easy to fabricate, and has the potential to detect biomolecules and chemical molecules at trace levels.



■ INTRODUCTION

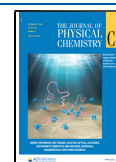
Photoinduced enhanced Raman spectroscopy (PIERS)^{1,2} extends the capabilities of surface-enhanced Raman scattering (SERS)^{3,4} by combining the plasmonic amplification of metal nanoparticles (MNPs) with the chemical enhancement provided by photoactivated semiconductor substrates. In SERS, the low cross section of Raman spectroscopy is hugely amplified by nanotextured surfaces (up to factors reaching 10¹⁰),^{5,6} which allows one to detect chemicals and biomolecular compounds in trace levels^{7–9} and, in suitable conditions, to reach single-molecule sensitivity.^{10,11} In PIERS, the sample is preirradiated with UV light in order to foster the injection of electrons (charge transfer, CT) from the semiconductor substrate to the MNPs.^{1,2,12,13} In both SERS and PIERS, the target molecules are detected after absorption (either chemisorption or physisorption) on nanostructured surfaces which provide a two-fold amplification effect; (i) a chemical enhancement based on the CT between the target molecule and the host surface (or conversely) that creates new energy levels yielding “resonant Raman” enhancement^{13,14} and (ii) an electromagnetic effect due to the field enhancement and confinement that the nanoscale texture of the hosting surface creates, and that, in turn, amplifies both the excitation and the Raman fields.^{15–17} SERS mainly exploits MNP substrates because these provide strong and locally concentrated electromagnetic fields when collective electron oscillations,

that is, localized surface plasmon resonances (LSPRs), are resonantly excited. In MNPs, in fact, the electromagnetic enhancement (10⁶ to 10⁸) overwhelms the chemical one (10² to 10³).⁷ In the last decade, a number of alternative materials, including semiconductors, dielectrics, and core–shell hybrids have shown the capability to provide SERS.^{6,18} While some nanomaterials, such as SiO₂@Au, SiO₂/TiO₂, or Si, are more effective in trapping and amplifying the electromagnetic field,^{18,19} some others, such as graphene²⁰ and layered transition-metal dichalcogenides (TMDs)^{21,22} base their SERS activity on CT.^{23,24} TMDs have the general formula MX₂, where M is a transition-metal atom and X is a chalcogen atom, and include molybdenum disulfide (MoS₂), tungsten disulfide (WS₂), molybdenum diselenide (MoSe₂), and molybdenum ditelluride (MoTe₂). Two-dimensional (2D) layered MX₂ (2DMX₂) has a crystal structure, three atoms thin, with the transition-metal atom sandwiched between the two chalcogens (X–M–X). High-purity single-layer 2DMX₂ can be produced by chemical vapor deposition,²⁵ whereas

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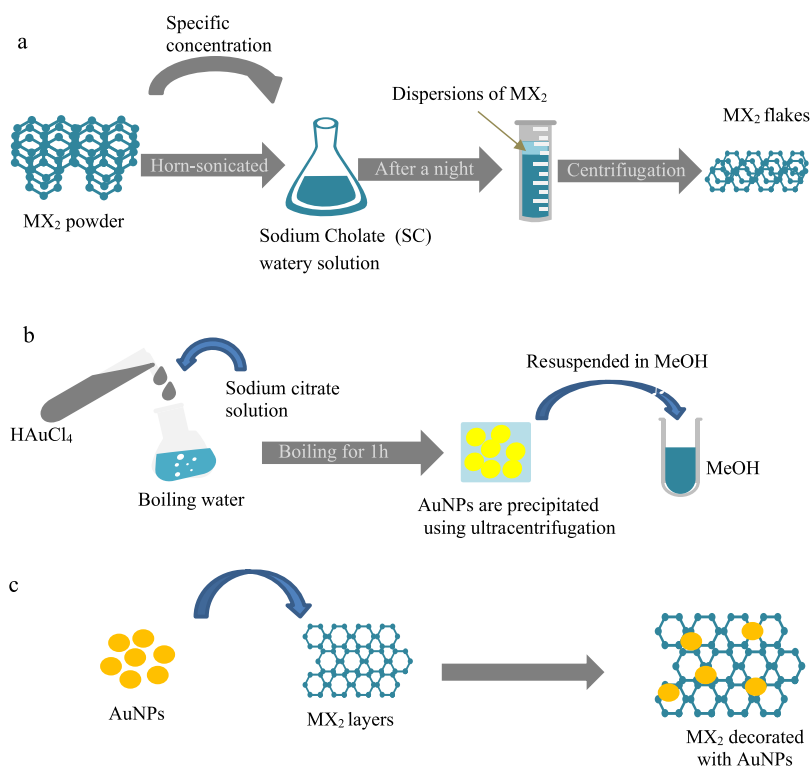


Figure 1. Schematic illustrations of the synthesis of (a) WS₂ nanosheets, (b) Au MNPs, and (c) Au@WS₂ nanocomposites.

polydisperse flakes with thickness varying from monolayer to ~ 10 layers and widths of few hundreds of nanometers are obtained by liquid-phase exfoliation (LPE), a low cost, scalable mass production technique.^{26–28} Because of their high conduction and catalytic activity, 2DMX₂ have been proposed as sensors.^{29,31} Hybrid 2DMX₂–graphene and 2DMX₂–MNPs have been shown as new materials for SERS,^{32–34} which can feature unique properties in which coupled plasmon-exciton states can be resonantly driven to trigger surface-enhanced resonant Raman scattering effects.³⁵

High-band gap semiconductors like titania^{1,2} and diphenylalanine peptide nanotubes,¹² decorated with Au and Ag nanoparticles have shown up as efficient PIERS substrates, which require photoirradiation in the UV. 2DMX₂ have been identified as potential substrates for PIERS because of the high carrier mobility, surface-to-volume ratio, and broadband absorption.²² No experimental proof of PIERS on 2DMX₂ has been reported, to our best knowledge.

Here, we show that tungsten disulfide (WS₂) nanosheets fabricated by LPE and decorated with gold nanoparticles (AuNPs) (Au@WS₂) can be used as efficient PIERS substrates. Proof-of-principle detection of 4-mercaptobenzoic acid (MBA) is shown at 10 μM concentration, with an overall PIERS signal enhancement of $\sim 10^6$, and the PIERS shows four times stronger signal than that of SERS.

METHODS

Reagents and Materials. Tungsten(IV) disulfide powder (particle size 2 μm , 99%, part no. 243639), sodium cholate hydrate (part no. C6445), gold(III) tetrachloride trihydrate (HAuCl₄·3H₂O), MBA (C₇H₆O₂S) (99%), sodium citrate dihydrate (>99%), and methanol (MeOH) (>99.9) were purchased from Sigma-Aldrich.

All chemicals were used as received, without further purification. All solutions were prepared with deionized water from a Millipore system.

MBA is used as a probe molecule for both PIERS and SERS to check the sensitivity of the Au@WS₂ substrates. AuNPs washed in DI water followed by respread in methanol to optimize the distribution of the nanoparticles while drying on the substrate.

Preparation of the WS₂ Nanosheets, AuNPs, and Au@WS₂ Nanocomposites. WS₂ nanosheets were prepared by LPE by following a procedure similar to those reported in refs.^{30,36,37} (see the sketch in Figure 1a). WS₂ powder at a concentration of 5 mg/mL was added to a watery solution of sodium cholate prepared at a concentration of 1.5 mg/mL. The solution was horn-sonicated for 30 min in an ice bath and then subjected to centrifugation (1500 rpm) for 1.5 h. The solution was decanted in a flask overnight. The supernatant, rich in few layers of nanosheets, was pipetted and stored away (inset of Figure 1), resulting in a dispersion, stable for months.

AuNPs were synthesized using the standard Turkevich–Frens procedure (see the sketch in Figure 1b).³⁸ HAuCl₄ (120 mg) was dissolved in 250 mL of boiling water. A 1% (w/w) sodium citrate solution (25 mL) was added and the reaction kept boiling for 1 h. AuNPs were precipitated by ultracentrifugation at 12,000g and resuspended in MeOH. The resulting solution features Au particles of an average size of 27 ± 5 nm at a concentration of $\sim 4 \times 10^{-10}$ M determined by UV–vis.³⁹ For SERS experiments, the solution was drop-cast onto glass substrates, followed by air drying, to give a surface coverage of ~ 250 AuNPs μm^{-2} .

Au@WS₂ nanosheets were prepared by mixing 100 μL of WS₂ solution with 100 μL of AuNP colloidal solution (Figure 1c). The mixture was stirred for 5 min. The PIERS substrates

are prepared by drop-casting of 50 μL on a glass slide, followed by air drying.

Apparatus and Measurements. A Branson 250 horn sonifier (power 250 W) was used for WS_2 LPE. Solutions were kept in an ice bath in order to prevent overheating.

The WS_2 dispersions were characterized by UV–vis (PerkinElmer Lambda 20) and Raman spectroscopy (Renishaw InVia) to determine the size and number of the average flakes of layers. For UV–vis, 1 cm length sample cells were employed. Raman spectra were acquired with a 633 nm He–Ne laser excitation (power 1.9 mW, spot size 4.4 μm^2). The instrument was coupled to a microscope using a 50 $\times$ objective for excitation and backscattering collection.

The Au@WS_2 dispersions were characterized by UV–vis spectroscopy, which was performed using a PerkinElmer Lambda 25 UV–vis for colloid nanoparticles in solution.

SERS and PIERS spectroscopy were carried out using a Renishaw InVia microRaman spectrometer (50 $\times$ objective, 1.9 mW laser power on a 4.4 μm^2 spot). For direct comparison between PIERS and SERS spectra, normalization of the Raman spectra was carried out using the standard deviations of the baseline signal for corresponding powder, SERS, and PIERS spectra, within the range of 1600–1800 cm^{-1} . UV preirradiation of the PIERS substrates was carried out using a UV-C (254 nm) light (2×8 W bulbs at a distance of 13 cm).

Transmission electron microscopy (TEM) images were obtained using a high-resolution transmission electron microscope Jeol 2100 with a LaB 6 source operating at an acceleration voltage of 200 kV. Micrographs were recorded on a Gatan Orius charge-coupled device.

RESULTS AND DISCUSSION

Nanomaterial Metrology. UV–vis spectroscopy provides information on flakes size and concentration of the WS_2 dispersions. When the light goes through a liquid dispersion of particles, its intensity in the forward direction is attenuated by both absorption and scattering, a phenomenon called extinction. For an optically thin dispersion, in which multiple scattering can be neglected, the intensity of the light transmitted through a cell of thickness l , containing particles at concentration c is given by $I_T = I_0 \exp[-\epsilon(\lambda)cl]$, where I_0 is the incident light intensity, and $\epsilon(\lambda)$ is the wavelength-dependent extinction coefficient. Commercial instruments measure the log in base 10 of the transmittance $T = I_T/I_0$, that is, the quantity $\text{ext}(\lambda) = \epsilon(\lambda)c = -[\log_{10} T(\lambda)]/l$ over a standard sample length ($l = 1$ cm typically), that is, a quantity that has the dimensions of an inverse length (1/cm).

The extinction spectrum of the WS_2 solution (Figure 2) shows the main features observed in LPE solutions of few-layer flakes²⁷ collected at room temperature under 254 nm excitation. These include the “D” band located at 337 nm due to the confinement of carrier.⁴⁰ Moreover, the absorption peaks A and B are obtained at 629 and 531 nm,^{27,41} which arise from direct excitonic absorptions at the K point of the Brillouin zone. We observed the C band at 468 nm due to optical transitions between the density of states peaks in the valence and conduction bands.⁴¹

We have used the second derivative method to determine all peak positions and to avoid fictitious shifts induced by the scattering background.⁵²

As the energy difference between A and C band’s frequencies is greatly sensitive to the number of layers, it can be used as a thickness indicator.⁴² According to Lan *et al.*,⁴²

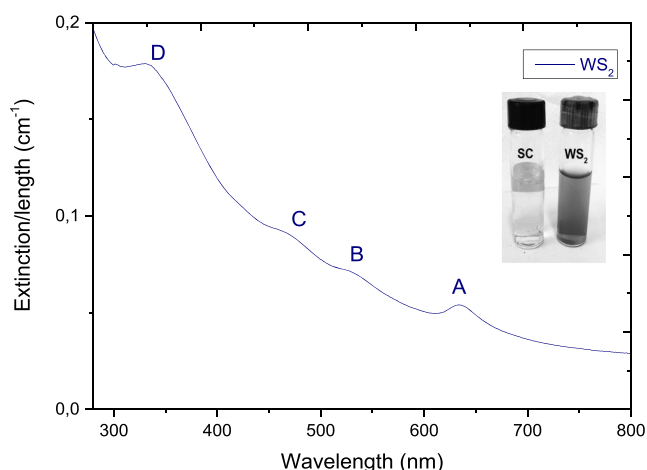


Figure 2. Extinction spectrum of the WS_2 solution used for experiments at room temperature. Inset: picture of a typical WS_2 dispersion in watery solution.

this difference is equal to 0.83 eV for WS_2 monolayers, while in our work, this difference is equal to 0.68 eV, proving that we have exfoliated a few layers of WS_2 . More precisely, we can use the position of the A band to determine the average number of layers, equal to $N \approx 8$ for our solutions, through the following empiric formula²⁷

$$N = 6.35 \times 10^{-32} e^{\lambda_A/8.51} \quad (1)$$

where λ_A is the position of the A-exciton in nm. An estimate of the average length $L \approx 100$ nm is obtained using the alternative metrics formula²⁷

$$L \text{ (in nm)} = \frac{1000 \text{ext}(A)/\text{ext}(290) - 7.6}{2.8} \quad (2)$$

valid for $L > 60$ nm and useful whenever an accurate determination of the extinction at the UV maximum (235 nm) is not possible, as in our case. A rough estimation of the concentration can be obtained from the extinction value at λ_A , considering that for flakes of size $L \approx 100$ nm, the extinction coefficient has been reported to be $\epsilon_A \approx 10 \text{ mL mg}^{-1} \text{ cm}^{-1}$.²⁷ We calculate a concentration $c = \text{ext}(A)/\epsilon_A \approx 5 \mu\text{g/mL}$ for our solutions.

UV–vis spectroscopy is used to determine the LSPR band of the MNPs (Figure 3a), which for small gold particles, is typically in the visible range.⁴³ Our particles show a single peak located at 521 nm, typical of a spherical morphology with a radius < 50 nm.⁴⁴

As the LSPR is not sensitive to the exact size for tiny particles,⁴⁵ we have used TEM to check the morphology. Figure 3b shows a typical TEM image of our AuNPs. The size distribution histogram in Figure 3c shows a radius of 27 ± 5 nm.

Raman Spectroscopy. Raman spectroscopy is an election tool to determine the vibrational modes in order to identify the crystalline structure of 2D layered materials. Furthermore, it is used to estimate the number of layers in the composition of the exfoliated 2D-MX₂.⁴⁶ Bulk WS_2 has an indirect band gap at 1 eV and belongs to the D_{6h} point group symmetry. The lattice vibrations at the Γ point can be expressed, therefore, as a sum of irreducible representations of the D_{6h} group⁴⁶

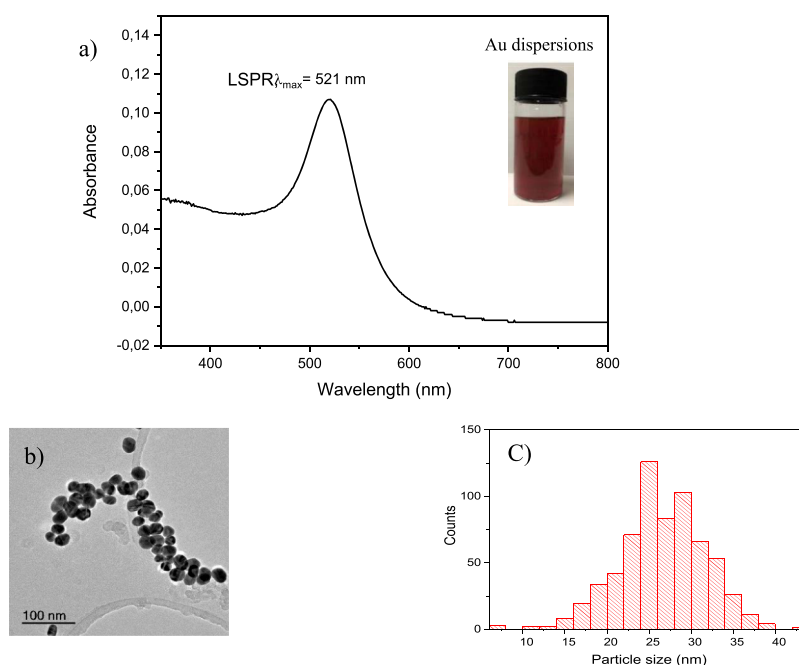


Figure 3. Characterization of AuNPs, (a) UV-vis absorption spectra of AuNPs. The LSPR absorption peak was observed at 521 nm. The data were obtained at room temperature using deionized water as blank. Samples were dispersed in deionized water. (b) TEM images of AuNPs and (c) histogram of particle size distribution of AuNPs.

$$\Gamma = A_{1g} + 2A_{2u} + 2B_{2g} + B_{1u} + 2E_{1u} + E_{1g} + 2E_{2g} + E_{2u} \quad (3)$$

Here, A_{1g} , E_{1g} , and E_{2g} are Raman optically active modes, A_{2u} and E_{1u} are acoustic modes, A_{2u} and E_{1u} are infrared active modes, and B_{2g} , B_{1u} , and E_{2u} are optically inactive phonon modes. In the monolayer/few layer form, the band gap changes from indirect (1 eV) to direct (2.1 eV),⁴⁷ and the symmetry point group changes from D_{6h} to D_{3d} if the number of layers (N) is even and to D_{3h} if N is odd. The irreducible representations change as follows⁴⁶

$$\Gamma_{\text{odd}} = [(3N - 1)/2](A'_1 + E'') + [(3N + 1)/2](A'_2 + E')$$

$$\Gamma_{\text{even}} = [3N/2](A'_1 + E'' + A'_2 + E') \quad (4)$$

The Raman spectrum measured at room temperature under a laser line of 633 nm (~ 1.96 eV) on our WS₂ is presented in Figure 4, where the sharp Raman peaks reflect good crystallization of the exfoliated WS₂ nanosheets. In Table 1, we report the list of all observed modes compared with the literature.

The spectrum is dominated by the out-of-plane A_{1g} mode at 420 cm⁻¹ and the 2LA(M) mode at 351 cm⁻¹. This is an acoustic mode considered as strong second-order resonance⁴⁸ overtone of the zone edge LA(M) mode at 174 cm⁻¹, caused by the disorder in the lattice.⁴⁹ The 2LA(M) mode overwhelms the E_{2g}^1 mode at 354 cm⁻¹ related to the in-plane vibration between the S and W atoms (inset of Figure 4). The E_{2g}^1 vibrational mode can be investigated after the peak deconvolution by Lorentzian fitting (Figure 4). Some vibrational modes are combined citing the 2LA(M)– E_{2g}^2 (M) mode located at 296 cm⁻¹ that is mainly attributed to the strong resonance between the phonon and B-exciton Raman in WS₂ excited by the 633 nm laser line.⁵⁰ The peak located at 701

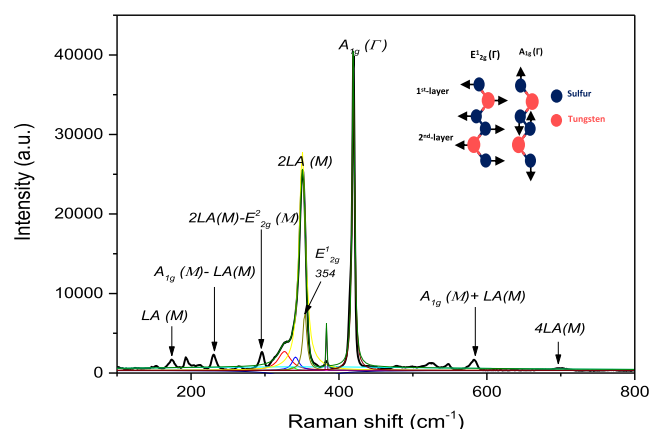


Figure 4. Raman spectrum of the WS₂ dispersion acquired at room temperature under the 633 nm laser line. The inset illustrates the atomic vibrations of the E_{2g}^1 and A_{1g} modes. The black curve is raw data, and the colored one is obtained with Lorentz fitting of the 2LA(M)– E_{2g}^1 band.

cm⁻¹ is assigned to the 4LA(M) mode, while the combination of A_{1g} with the LA mode ($A_{1g} \pm \text{LA(M)}$) is observed at 583 and 230 cm⁻¹, respectively. This combination is caused by the atomic displacements of A_{1g} in the c -axis direction.

As mentioned above, we can estimate the thickness (expressed in the number of layers) of the exfoliated MX₂ using the Raman data.⁴⁶ Indeed, the shift $\Delta\omega(A-E) = \omega(A_{1g}) - \omega(E_{2g}^1)$ is sensitive to the number of layers and can be used for metrology. However, it cannot be a way to find out the thickness in our work because the intensity of A_{1g} (Γ) is much stronger than that of 2LA(M).⁵⁰ Consequently, we have estimated the WS₂ thickness through the relation $\Delta\omega(A-2LA) = \omega(A_{1g}) - \omega(2LA)$ according to Li *et al.*,⁵⁴ which is around 69 cm⁻¹, the same difference as in our case, indicating that on average, the WS₂ nanosheets are composed of >4 layers. For

Table 1. Raman Mode Assignment of WS₂

modes	frequencies from this work (cm ⁻¹)	frequency (cm ⁻¹) from reference					
		50	49	51	52	53	31
LA(M)	174		176				
A _{1g} -LA(M)	230		230				
2LA(M)-E _{2g} ² (M)	296		297				
2LA(M)	351	350	352	350	353		
A _{1g} (Γ)	420	416.8	417			418	
E _{2g} ¹ (Γ)	354			355		357	
4LA(M)	701		704				705
A _{1g} + LA(M)	583		582				585

further confirmation of our calculation, we have used the intensity ratio between the 2LA and A_{1g} modes⁵⁰ to estimate the WS₂ thickness. We find $I_{2LA}/I_{A_{1g}} = 0.63$, indicating a few layers occurrence (for the monolayer, the ratio is ~ 5.6).⁵⁰ Finally, the quality of the WS₂ layers can be identified looking at the full-width at half maximum (fwhm) and intensity of the characteristic peaks in the Raman spectrum (A_{1g} and 2LA).⁵⁴ The fwhms are equal to 11 and 4 cm⁻¹ for 2LA(M) and A_{1g}(Γ), respectively, proving the good quality of the WS₂ flakes.

SERS and PIERS Response of the Au@WS₂ Nanosheets. SERS and PIERS are complementary techniques and differ by the preirradiation step. In our case, we have used MBA as an analyte to check the substrate's performances (see Supporting Information).⁵⁵ Figure 5 displays the SERS (dark

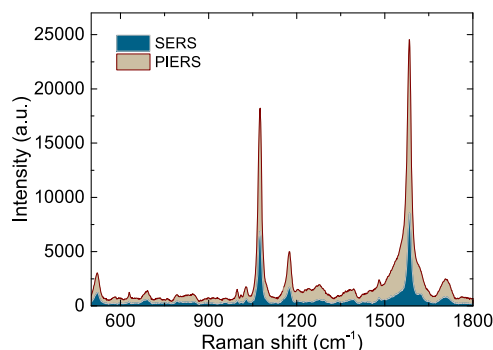


Figure 5. Room temperature SERS (dark cyan) and PIERS (light grey) spectra of MBA on Au@WS₂ nanosheets.

cyan) and PIERS (light gray) response of the Au@WS₂ nanosheets substrate excited at 633 nm at room temperature. The mode frequencies, compared to the normal Raman (see Figure S1, Supporting Information), are reported in Table 2.

The WS₂-AuNP SERS spectrum covers most modes detected in the normal Raman spectrum of MBA, indicating the same assignments with a red shift. This offset is caused by the structural deformation in the Au-conjugated SERS reporter molecule as a result of the binding event.⁵⁶ Consequently, it is contributed by a change in the polarizability of MBA-AuNPs, according to Ma *et al.*⁵⁷ Two main peaks are observed at 1584 and 1075 cm⁻¹. The first one is assigned to the ring breathing axial deformation mode, which is stronger than the intensities of the other bands. The second peak is the in-plane ring breathing mode.⁵⁸ This mode indicates the existence of a certain angle 'θ' between the plane of the benzene ring and the surface of AuNPs according to Ma *et al.*⁵⁹ The strength of the in-plane modes of MBA in SERS suggests that the analyte is

Table 2. Raman Mode Frequencies and Assignments of MBA Compared to the SERS/PIERS Modes of MBA Adsorbed on WS₂-AuNPs

normal Raman modes (cm ⁻¹)	SERS and PIERS modes (cm ⁻¹)	assignments
274		Au-S
	420	out-of-plane mode (A _{1g}) of WS ₂
344		C-H in-plane deformation, β(C-H)
	522	ring out-of-plane bending
632	633	OCO bending and C-S stretching, δ(O-C-O) & ν(C-S)
1098	1075	in-plane ring breathing mode
1183		C-O stretching, ν(C-O)
1453	1480	COO ⁻ stretching, ν(COO ⁻)
1595	1584	ring breathing axial deformation modes
1617	1706	C=O stretching, ν(C=O)

adsorbed on the surfaces of AuNPs through the S atom. Besides, the appearance of two extra peaks, which are located at 420 and 522 cm⁻¹ indicate the out-of-plane mode (A_{1g}) of WS₂ and the ring out-of-plane bending,⁵⁹ respectively. Besides, the Au-S, C-H in-plane deformation and C-O stretching modes disappeared.

We observe an enhancement of a factor ~ 4 in the PIERS spectrum over normal SERS, measured on the 1584 cm⁻¹ peak. We attribute this extra enhancement to charge transfer (CT),⁶⁰ that in turn induces a chemical enhancement because of the charge migration from the semiconductor substrate to the metallic NPs, which comes from the UV irradiation. Comparative measurements on an AuNP substrate (without WS₂) show no extra enhancement, thus, indicating that amplification occurs only in the presence of the semiconductor.

The enhancement factor in SERS or PIERS can be determined as^{1,5,61}

$$AEF = \frac{I_{PIERS(SERS)}/C_{PIERS(SERS)}}{I_{Raman}/C_{Raman}} \quad (5)$$

This factor is known as the analytical enhancement factor (AEF), where $I_{PIERS(SERS)}$ and I_{Raman} are the scattering intensities of the average PIERS(SERS) and normal Raman scattering, respectively, normalized to the power and integration time. $C_{PIERS(SERS)}$ and C_{Raman} are the analyte concentrations, respectively. Here, we estimate an $AEF_{SERS} \approx 2 \times 10^5$ and an $AEF_{PIERS} \approx 9 \times 10^5$.

To sum up, a sketch is presented in Figure 6 for the happening phenomena, which occur upon the UV-C irradiation in the PIERS technique. Here, the UV-C irradiation

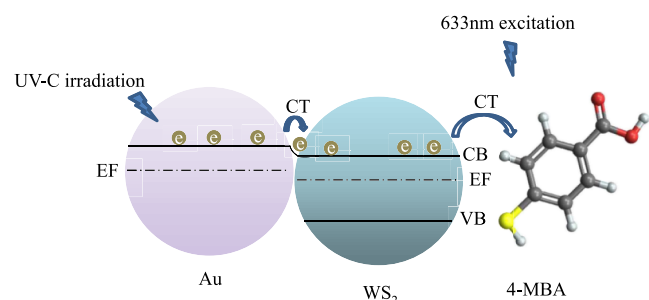


Figure 6. Sketch presenting the phenomena happening upon the UV-C irradiation in the PIERS technique.

used in the preirradiation step in the PIERS technique is used to bombard the LSPR, creating hot spots that will contribute to the signal enhancement since the excitation energy is similar to the AuNP Fermi energy producing the CT between the AuNPs and the semiconductor.

Photoirradiation of the Au@WS₂ Composites. Figure 7 shows the extinction spectra acquired on the Au@WS₂

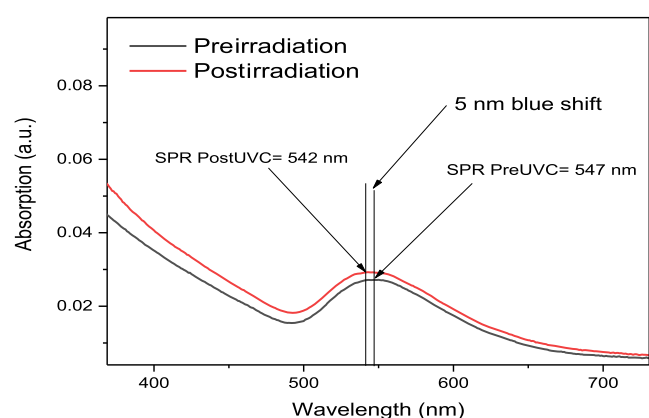


Figure 7. Extinction spectra of Au@WS₂ composites before (black line) and after (red line) irradiation with UV-C light at room temperature.

substrate pre- and post-UV-C light irradiation (black and red lines). The LSPR peak is observed at 547 nm, red-shifted for what is expected for individual 27 nm AuNPs (520 nm), indicating some degree of aggregation upon absorption on the surface of the WS₂ nanosheet. The Au@WS₂ nanocomposites strongly absorb in the UV.

The blue shift of the LSPR (~5 nm) is interpreted as the activation of the AuNPs as a consequence of the increase of electron density due to a WS₂ to AuNP CT. The electron density variation $\Delta N/N$ can be estimated as^{1,12}

$$\frac{\Delta N}{N} = -2 \frac{\Delta \lambda}{\lambda} \approx 1.8\% \quad (6)$$

where $\Delta \lambda$ measures the LSPR peak shift in the AuNPs@WS₂ nanocomposites.

WS₂ shows an indirect-to-direct gap crossover thinned to a single monolayer.⁴¹ CT between Au NPs and MX₂ has been observed in the literature by PL quenching of MoS₂.⁶² The similarities between MoS₂ and WS₂ suggest an analogous CT process between AuNPs and WS₂. The displacement of the charges can be identified according to semiconductor theory.¹³ The contact between a semiconductor and a metal can either be an Ohmic contact or a Schottky contact, depending on the

work function's sign of both the semiconductor and metal.¹³ In fact, if the work function of the semiconductor is greater than the metal one, electrons will flow from the metal to the semiconductor, and the contact is then Ohmic.¹³ If the work function of the semiconductor is lower than the metal, we have a Schottky, where a positive space charge is created in the junction, that is, electrons flow from the semiconductor to the metal, creating a Schottky barrier that will block the CT.¹³ In our framework, the contact between the semiconductor and the noble metal is Ohmic, where the high concentration of the electron will produce high conductivity in the junction.¹⁸ Indeed, the Fermi level of multilayer WS₂ has been reported to be 5.89 eV⁶³ (that of single layer is equal to 4.6 eV),⁶⁴ while for Au it is 5.1 eV.⁶² The 0.79 eV energy level offset results in a band bending in the Au@WS₂ hybrid nanosheet. Upon illumination, the electrons in the excited states of AuNPs transfer to WS₂; see Figure 8.

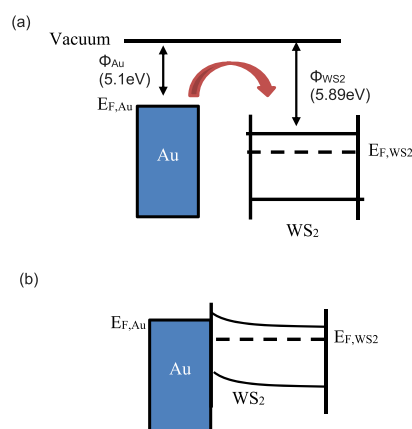


Figure 8. CT mechanism in Au@WS₂ nanoflakes. (a) Energy band diagram for AuNP and WS₂ nanosheets, showing the relative positions of Fermi levels with respect to the vacuum level. The arrow represents the transfer of electrons from Au to WS₂ after the contact is established. (b) Energy band diagram of Au@WS₂ nanoflakes showing band bending after establishing the contact between Au and WS₂.

CONCLUSIONS

In conclusion, we have proved that Au@WS₂ nanosheets can be used as efficient PIERS substrates for molecular detection. The WS₂ nanosheets were prepared by LPE with good crystallinity, featuring an average number of layers $N \approx 8$ and a mean flake size ≈ 100 nm. The PIERS-active substrates have been prepared by mixing and casting of the WS₂ nanosheets with AuNPs grown by chemical means. Proof-of-principle PIERS experiments have been carried out by detecting MBA molecules at 10^{-5} M concentration. PIERS spectra on Au@WS₂ substrates irradiated with UV-C light evidenced a Raman signal of MBA four times stronger than SERS from Au@WS₂ substrates without UV irradiation, a phenomenon which is not observed on substrates made of simple AuNPs. The PIERS extra enhancement is attributed to a chemical enhancement due to metal-to-semiconductor CT. Such a CT is independently proved and quantified to ca. 1.8%, forming the blue shift observed in the AuNP LSPR upon UV irradiation. The global PIERS enhancement factor is estimated to be of the order 10^6 . The simplicity of fabrication of our samples, associated with

the versatility and low cost makes these materials very promising for sensing applications in different areas.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.0c04664>.

Raman spectrum of MBA (PDF)

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